

Control of ROS-based Mobile Robots: From Collision-Risk Potentials to Velocity-Obstacle Planning in Simulation

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Abstract—This paper presents the advanced simulation and software in loop phase of our project *Control of ROS-based Mobile Robots*. Building on the previously implemented pipeline of (i) Collision-Risk Potential Fields (CRPF) for risk quantification, (ii) Velocity Obstacles (VO) for long-horizon safety, and (iii) the Dynamic Window Approach (DWA) for constraint enforcement, we focus on cooperative agent behavior and realistic physics. We introduce a new coordinated yielding scenario where a trailing subject vehicle (SV) detects a merging obstacle and modulates its speed to facilitate a safe lane change for the obstacle vehicle. Furthermore, we transition our entire control stack from MATLAB-based kinematics into a ROS-integrated Gazebo environment, incorporating the exact physical and inertial parameters of ROS based Turtlebots Lab. The resulting simulations confirm that our risk-aware logic remains effective when subjected to realistic dynamics and sensor constraints, providing a verified foundation for the upcoming deployment on physical hardware.

I. INTRODUCTION

In the previous semester, we focused on the mathematical foundation and basic reproduction. We implemented the CRPF equations and tested them in four initial cases: a single subject vehicle (SV) avoiding stationary/moving obstacles, and a 4-vehicle platoon doing the same. These tests were done in a MATLAB environment using kinematic models.

This semester, we have expanded our work in two major ways. First, we introduced a more complex multi-agent scenario called **Coordinated Yielding**. In this case, instead of just bypassing an obstacle, two SVs must work together to create a gap for an obstacle vehicle to pass through for its lane change. Second, we transitioned our entire simulation stack from MATLAB kinematics into the ROS-integrated Gazebo engine. By using the actual physical and inertial parameters of the laboratory TurtleBot3 robots, we are now testing how our algorithms handle realistic dynamics and sensor constraints.

Our specific goals for this phase were:

- **Scenario Expansion:** Implement a coordinated yielding case where a trailing SV modulates its speed to let an obstacle vehicle (OV) change its lane through gap.
- **Transition to ROS/Gazebo:** Migrate the control nodes from MATLAB to ROS 2, ensuring the logic works in a physics-based environment.
- **Physical Parameter Matching:** Configure the Gazebo simulation with exact TurtleBot Burger parameters (mass,

velocity limits, and friction) to prepare for lab deployment.

II. BACKGROUND AND RELATED WORK

A. Artificial Potentials and CRPF

Classical *artificial potential fields* (APF) penalize proximity to obstacles and attract a robot to goals [?]. APF is intuitive but struggles with dynamic obstacles and has local minima issues. CRPF augments APF by injecting *semantics* (vehicle mass, type, speed, acceleration, road factors) and *directionality*; its equipotentials elongate along the obstacle’s motion direction (akin to a Yukawa-type anisotropic field [?]), which better reflects risk distribution for moving obstacles.

B. Velocity Obstacles (VO)

VO marks the set of robot velocities that will inevitably lead to collision with an obstacle within a horizon [2]. By excluding VO, the robot limits itself to safe velocities. VO is appealing because it lifts a complex geometric time-evolution problem into a simple instantaneous constraint in velocity space.

C. Dynamic Window Approach (DWA)

DWA is a short-horizon search over velocities reachable by the robot under bounded accelerations [3]. It rolls out candidate commands for a brief lookahead and chooses the best by a cost that balances safety and progression. DWA is lightweight and compatible with on-board computation.

D. Hybrid Automata and MPC

Hybrid automata glue discrete maneuvers (cruising vs. platooning) to continuous dynamics [?]. While Wang *et al.* combine CRPF/VO with MPC for longer-horizon tracking [?], our baseline simulations use DWA to keep runtime small while reproducing the paper’s logic (risk-aware pruning followed by local optimization).

III. QUANTIFYING THE CRPF

CRPF assigns a *risk magnitude* around an obstacle and a *risk direction* pointing outward from the obstacle’s center. The field used in [1] (their Eq. (11)) can be written as

$$\mathbf{E}_V(\mathbf{p}_s) = \frac{G M(v_o) T(\kappa, s) \phi(\mathbf{a}_o, \hat{\mathbf{r}})}{\|\mathbf{r}_d\|^\zeta} \hat{\mathbf{r}}, \quad (1)$$

with virtual mass

$$M(v_o) = m (1.566 \times 10^{-14} v_o^{6.687} + 0.3354), \quad (2)$$

type/size factor

$$T(\kappa, s) = \left(\frac{s}{s^*}\right)^{\omega_1} \left(\frac{\kappa}{\kappa^*}\right)^{\omega_2}, \quad (3)$$

acceleration factor

$$\phi = \frac{K}{K - \|\mathbf{a}_o\| \cos \vartheta_a}, \quad \cos \vartheta_a = \frac{\mathbf{a}_o}{\|\mathbf{a}_o\|} \cdot \hat{\mathbf{r}}, \quad (4)$$

and the anisotropic pseudo-distance

$$\mathbf{r}_d = \begin{bmatrix} \varepsilon(x_s - x_o) e^{-\rho v_o} \\ \varepsilon(y_s - y_o) \end{bmatrix}, \quad \zeta > 0, \rho \geq 0. \quad (5)$$

Faster/heavier obstacles and positive acceleration toward the subject inflate risk; lateral offsets reduce risk when lanes separate agents.

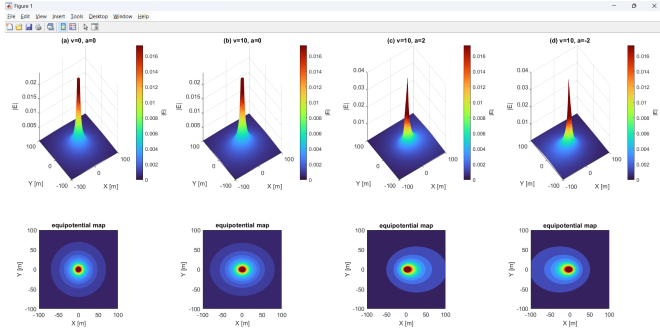


Fig. 1: CRPF reproduction: speed and acceleration alter the anisotropy of the risk field.

IV. VELOCITY OBSTACLES AND FEASIBLE VELOCITIES

A. VO derivation

Let $\mathbf{p}_s, \mathbf{p}_o$ be subject and obstacle positions; with disks of radii r_s, r_o , a collision occurs if the ray $\{\mathbf{p} + t\mathbf{v}\}$ (with relative $\mathbf{p}, \mathbf{v}$) intersects the Minkowski sum disk of radius $r_s + r_o$. The set of inducing subject velocities is the VO cone [2]. We also use an H -time VO to prune velocities that would collide within horizon H .

B. Dynamic Window and Reachability

Not all velocities are achievable instantly; the *dynamic window* collects velocities reachable in Δt under bounded linear and angular accelerations [3]. We (i) sample (v, ω) , (ii) roll out N steps, (iii) accumulate CRPF cost along the rollout, (iv) discard candidates that exit the lane or enter the collision disk, and (v) choose the minimal-cost candidate. VO is used *before* rollout to drop candidates that are guaranteed to collide; the CRPF cost then shapes preferences among the safe ones.

V. VEHICLE MODEL AND COST FUNCTIONS

We use the unicycle kinematics:

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega. \quad (6)$$

Single-vehicle cost:

$$J = \alpha w_r \overline{E_V} + w_v (v_{\text{des}} - v)^2 + w_\omega \omega^2 + w_{\text{lat}} |\overline{\text{CTE}}| + w_{\text{head}} |\overline{\psi}| + w_{\text{end}} |\text{CTE}_{\text{end}}|. \quad (7)$$

For platoons (followers) we add headway and relative-speed terms:

$$J_{\text{platoon}} = J + w_{\text{gap}} |d - d^*| + w_{\text{rel}} |v - v_{\text{pred}}|. \quad (8)$$

We gate the risk term with $\alpha \in [0, 1]$ based on $E_V \in [E_{\text{lo}}, E_{\text{hi}}]$ (paper's “yellow band”).

Extensions for Platoon Scenarios (Split/Merge)

We adopt a light hybrid automaton with discrete modes

PLATOON $\leftrightarrow$ CRUISE.

Split is triggered if $\text{TTC}_{\text{OV}} < \tau_1$ or $\text{TTC}_{\text{pred}} < \tau_2$ or $E_V \geq E_{\text{hi}}$. **Merge** is permitted if $E_V \leq E_{\text{lo}}$, the headway to the predecessor is $d \in [d^* - \delta, d^* + 2\delta]$, and the vehicle is back on MAIN.

Lateral motion is an FSM over lane centers $\{\text{LEFT}, \text{MAIN}, \text{RIGHT}\}$. Lane changes use a quintic progress $s(u) = 10u^3 - 15u^4 + 6u^5$ with $u = t/T_{\text{LC}}$. While on a passing lane and still behind the OV, a pass-floor $v \geq v_{\text{OV}} + \Delta v$ is enforced. After clearing the OV by Y_{clear} , a *recenter lock* holds the target at MAIN for a few ticks to complete the maneuver cleanly. This mirrors the discrete maneuver completion logic of [1].

VI. SIMULATION SETUP & RESULTS

Case 1: Single SV vs. stationary OV

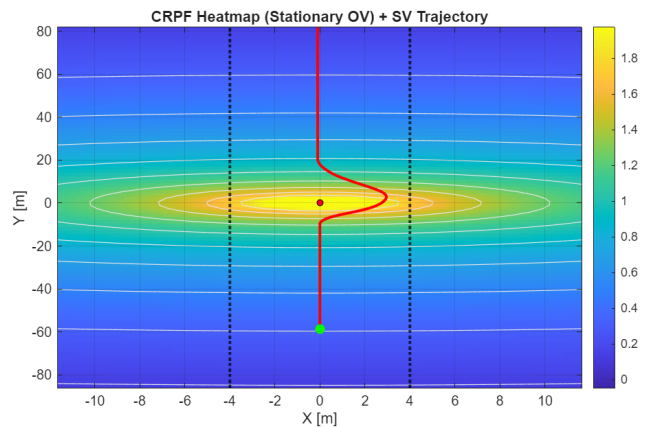


Fig. 2: Case 1—CRPF heatmap (stationary OV) with white contours and the SV trajectory (red).

Explanation. The SV accelerates toward v_{des} and bends smoothly to skirt the CRPF contours while staying inside ± 4 m bounds. After clearing Y_{clear} the recenter lock snap the vehicle

back to the centerline, completing a symmetric avoid-recenter maneuver without oscillations. TTC remains above the safety threshold for the entire horizon, and the minimum center-to-center distance stays comfortably larger than $r_s + r_o$. Gated CRPF (α) prevents premature lateral drift, while VO pruning removes long-horizon unsafe candidates so DWA evaluates only feasible, low-risk rollouts.

Case 2: Single SV vs. moving OV (constant speed)

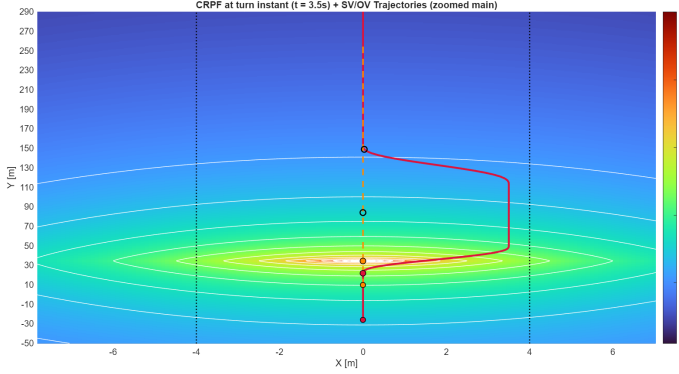


Fig. 3: Case 2—CRPF recomputed at the *turn instant*; start/turn markers for SV and OV.

Explanation. The time-aligned CRPF shows elongated contours in the OV motion direction. The SV initiates a lateral move exactly as the translated risk peak encroaches, then merges back after Y_{clear} . Compared to Case 1, the turn occurs slightly later and farther downstream because the risk hill itself is moving. The pass-speed setpoint keeps TTC above the safety threshold while the SV is still behind, and the minimum distance remains well above $r_s + r_o$. Gated CRPF avoids premature drift, VO pruning removes long-horizon unsafe candidates, and the turn-instant anchoring prevents misinterpretation that would arise from overlaying a $t=0$ field on a moving-OV scenario.

Case 3: Platoon (4 SVs) vs. stationary OV (lane-restricted split/merge)

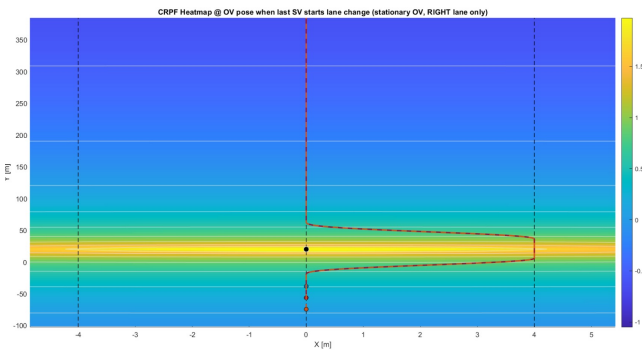


Fig. 4: Case 3—Anchored CRPF with all four SV trajectories and lane-restricted overtake.

Explanation. Split events occur only for vehicles approaching unsafe TTC or elevated CRPF; lane availability windows prevent side conflicts. The trajectories show clean LEFT/RIGHT passes and synchronized merges. Headway time series recover toward d^* after merges, confirming that the lightweight platoon cost stabilizes gaps without inducing oscillations.

Case 4: Platoon (4 SVs) vs. moving OV (cruising)

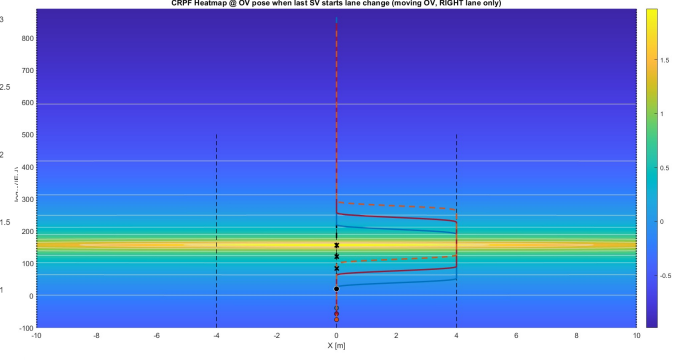


Fig. 5: Case 4—Anchored CRPF with trajectories during moving-OV cruising.

Explanation. The moving CRPF hill shifts split timing downstream relative to Case 3. The pass-speed floor prevents prolonged side-by-side states and shortens exposure in passing lanes. Post-merge headways settle near d^* , confirming the merge guard and recenter lock produce stable “cruise-as-platoon” behavior after the maneuver.

Case 5: 2-SV Formation vs. Lateral OV Crossing (Dynamic Gap Windowing)

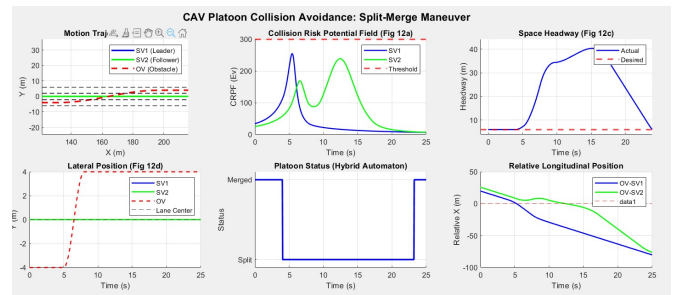


Fig. 6: Case 5—Analysis of splitting and merging.

Below are simulation's instances

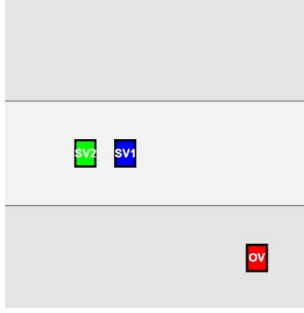


Fig. 7: $t = 0$



Fig. 8: OV proceeds to change the lane to right, SV platoon starts splitting.



Fig. 9: OV is in the right lane now.

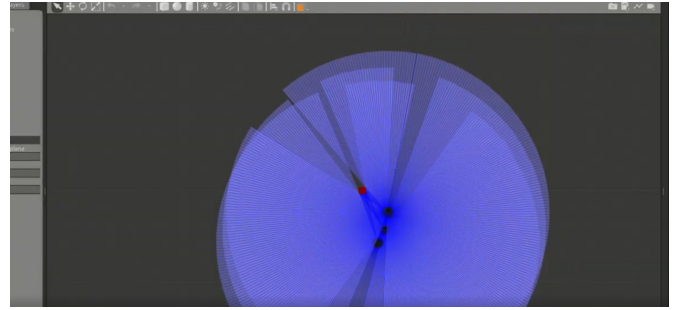
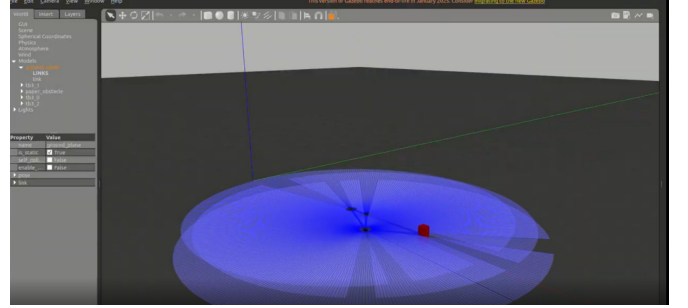
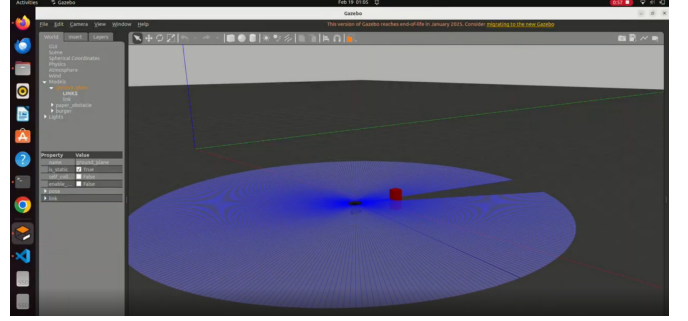


Fig. 10: SV2 merges again to form a platoon and OV left behind cause of lesser speed.

Explanation. There is a platoon of 2 subject vehicles going in the MIDDLE lane of the road at some more speed then Obstacle vehicle which is on RIGHT lane. The Obstacle vehicle wants to change its lane from RIGHT to LEFT lane goind via MIDDLE lane. But the speed of both SVs and OVs and separation between them is chosen in such a way that only front SV has a window to overtake, hence changing the mode from platooning to cruising and rear SV slows down. This creates a gap which lets OV to pass through and later rear SV catches up.

VII. GAZEBO SIMULATIONS SNAPSHOTS

Below are some of our screenshots attached to showcase how they look.



VIII. UPCOMING PHASE GOALS

Future work. Integrating them on mobile-based bots for platooning experiments.

IX. CONCLUSION

So far we have reproduced the core perception–planning loop of [1] in five scenarios. We are also done with simulating 4 of them on gazebo with physical parameters with ROS.

REFERENCES

- [1] S. Wang, Z. Li, B. Wang, and M. Li, "Collision Avoidance Motion Planning for CAV Platoon Merging and Splitting with a Hybrid Automaton Architecture," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 2, pp. 1445–1459, 2024.
- [2] P. Fiorini and Z. Shiller, "Motion Planning in Dynamic Environments Using Velocity Obstacles," *International Journal of Robotics Research*, vol. 17, no. 7, pp. 760–772, 1998.
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